Introduction to Programming

Lecture 4

VU
Virtual University

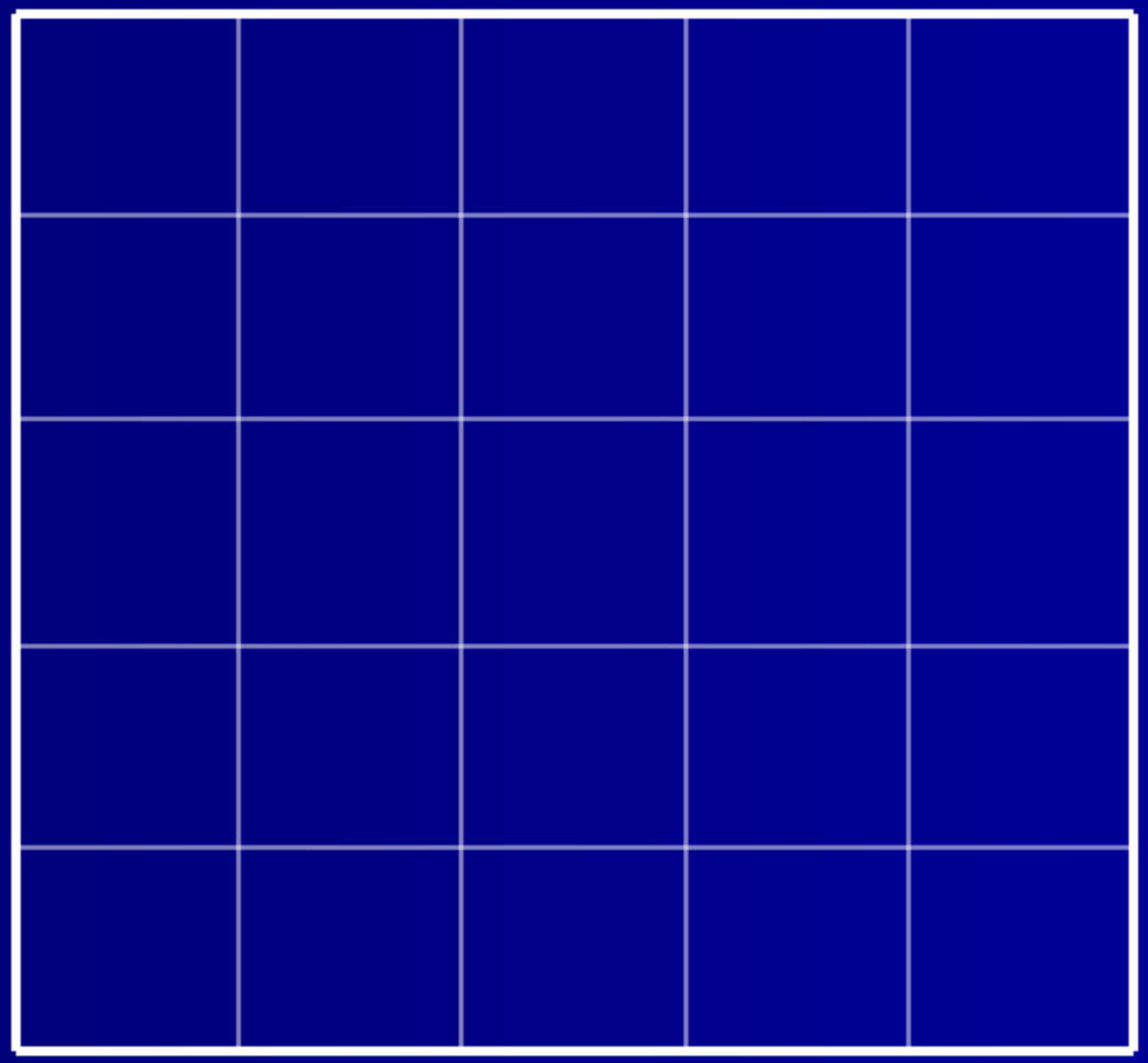
Key Words of C

- main
- if
- else
- while
- do
- for

Memory

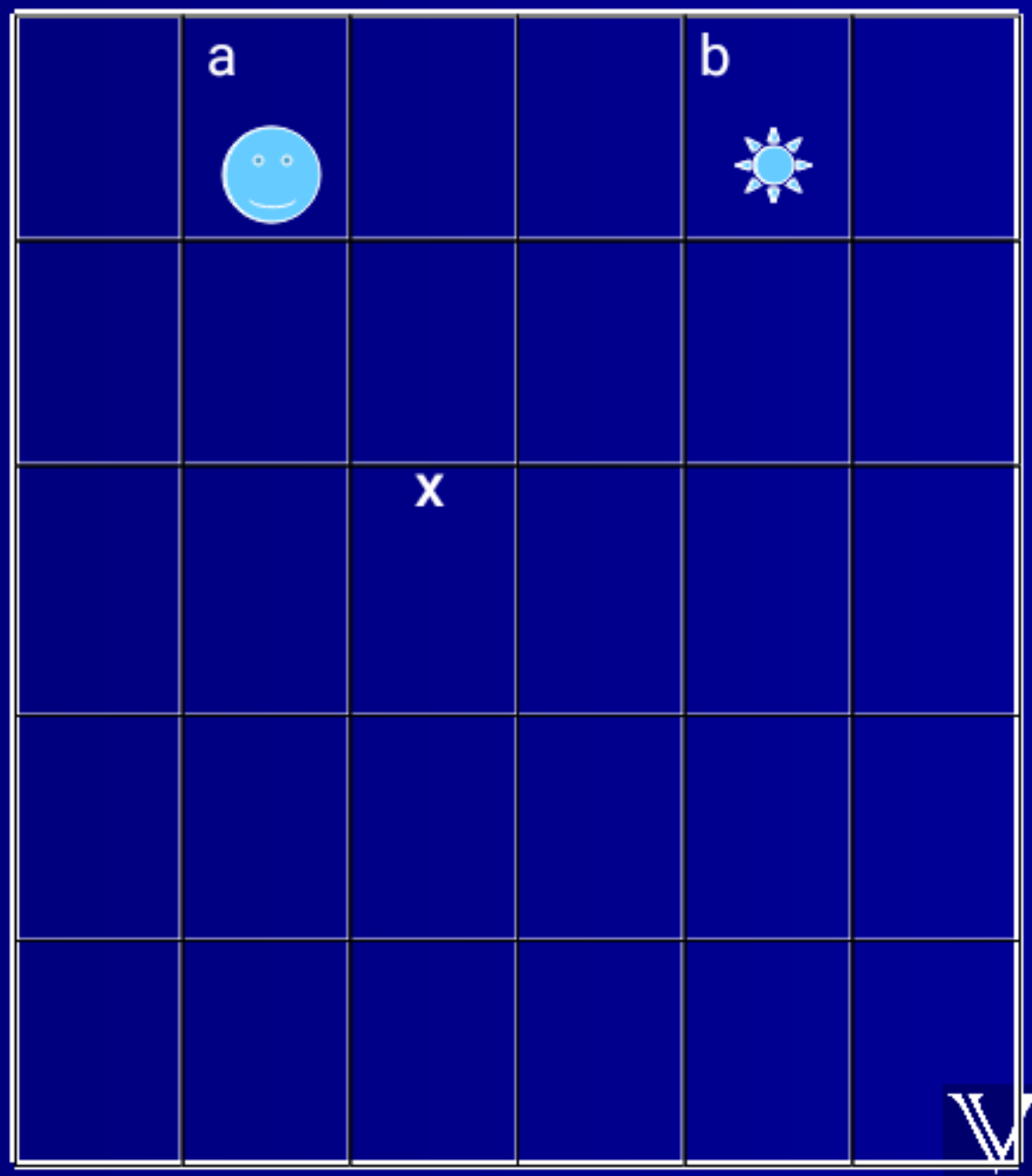
x = 2 + 4 ;

= 66



Memory

x = a + b ;



```
#include <iostream.h>
main ( )
{
 int age1, age2, age3, age4, age5, age6, age7, age8, age9, age10 ;
 int TotalAge ;
     int AverageAge ;

 cout << " Please enter the age of student 1: " ;
 cin >> age1 ;

 cout << " Please enter the age of student 2: " ;
 cin >> age2 ;
 :
 :
 TotalAge = age1+ age2 + age3+ age4+ age5+age6+ age7+ age8+age9 + age10 ;
 AverageAge = TotalAge  / 10 ;

 cout<< "The average age of the class is :" << AverageAge ;
}
```

Quadratic Equation

- In algebra

$$y = ax^2 + bx + c$$

- In C

$$y = \underbrace{\underbrace{a*x*x}_{} + \underbrace{b*x}_{} + c}_{}$$

a*b%c +d

a*(b%c) = a*b%c

?

Discriminant

$$\frac{b2 - 2a}{4c}$$

= b*b - 4*a*c /2 *a Incorrect answer

Solution

= (b*b - 4*a*c) /(2 *a) Correct answer

- No expression on the left hand side of the assignment
- Integer division truncates fractional part
- Liberal use of brackets/ parenthesis

Interesting Problem

Given a four-digit integer, separate and print the digits on the screen

Analysis

- Number = 1234
- Take the remainder of the above number after dividing by 10
  Eg 1234 / 10 gives remainder 4
  1234 % 10 = 4
- Remove last digit
  - 1234/10 = 123.4
  - 123 (Truncation due to Integer Division)
- 123 %10 gives 3
- Remove last digit
  - 123/10 = 12.3
  - 12 (Truncation due to Integer Division)
- 12 % 10 gives remainder 2
- Remove last digit
  - 12/10 = 1.2
  - 1 (Truncation due to Integer Division)
- Final digit remains

Code

```
#include <iostream.h>
main ( )
{
        int number;
        int digit;
        cout << “Please enter a 4 digit integer : ”;
        cin >> number;
        digit = number %10;
        cout <<“The digit is: “ << digit << ‘\n’;        // first digit; and then << ‘\n’
        number = number / 10;
        digit = number % 10;
        cout <<“The digit is: “ << digit << ‘\n’;
        number = number / 10;
        digit = number % 10;
        cout <<“The digit is: “ << digit << ‘\n’;
        number = number / 10;
        digit = number % 10;
        cout <<“The digit is: “ << digit;

}
```

Special Character Newline

\n